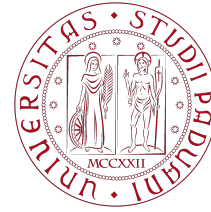


Final Report

Physics of Complex Networks: Structure and Dynamics



UNIVERSITÀ
DEGLI STUDI
DI PADOVA

Areas of physics by complexity



Newton's
Mechanics

Electro-
Magnetism

Special
Relativity

Quantum Mechanics
General Relativity

Quantum
Field Theory

Complexity
Science

Epidemics on temporal networks & the AS-level Internet from BGP

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1 | Epidemic spreading on temporal networks

Task leader(s): *Giancarlo Venturato*

1.1 | Models and mean-field thresholds

We study the four canonical compartmental models on *time-varying* contact networks. A node is Susceptible (S), Exposed (E, infected but not yet infectious), Infectious (I) or Removed (R). In discrete time, on every contact of snapshot t an S–I pair transmits with probability β ; an infectious node leaves state I with probability μ , and in SEIR an exposed node matures with probability σ . The four models are the same core loop restricted to different states [8]: SI (no recovery), SIS (I→S), SIR (I→R) and SEIR (S→E→I→R).

The order parameter is the endemic prevalence (SI/SIS) or the final epidemic size (SIR/SEIR), which is non-zero only above a critical ratio $(\beta/\mu)_c$. On the *activity-driven* temporal network the heterogeneous mean-field threshold is governed by the *activity* moments and not by the aggregated degree [9],

$$\left(\frac{\beta}{\mu}\right)_c^{\text{AD}} = \frac{1}{m(\langle a \rangle + \sqrt{\langle a^2 \rangle})}, \quad (1.1)$$

whereas treating the whole observation window as a single *static* graph gives the classical heterogeneous- (HMF) and quenched- (QMF) mean-field results evaluated on that aggregate,

$$\left(\frac{\beta}{\mu}\right)_c^{\text{HMF}} = \frac{\langle k \rangle}{\langle k^2 \rangle}, \quad \left(\frac{\beta}{\mu}\right)_c^{\text{QMF}} = \frac{1}{\lambda_{\max}(A)}. \quad (1.2)$$

Because the aggregate accumulates every contact ever made, $\langle k^2 \rangle$ and $\lambda_{\max}$ are large and the static threshold is orders of magnitude below Eq. (1.1): assuming that contacts are simultaneous systematically over-estimates spreading. Making this contrast explicit is the aim of the task.¹

1.2 | Synthetic temporal topologies and method

A temporal network is a sequence of T single-step snapshots. We use two structurally different synthetic generators, so the models are exercised across a variety of topologies.

¹The simulator uses discrete-time per-step probabilities (β, μ) , which map onto the continuous rates of Eqs. (1.1)–(1.2) for $\beta, \mu \ll 1$; the agreement in Fig. 1.1(b) confirms the mapping in the regime studied.

(i) The *activity-driven* (AD) model [9]: each node draws an activity $a_i \propto x^{-\gamma}$, $x \in [\varepsilon, 1]$; at every step it is active with probability a_i and, if so, fires m links to random partners. There is no fixed backbone. (ii) An *edge-activated* process on a fixed *scale-free* (Barabási–Albert) or *homogeneous* (Erdős–Rényi) backbone whose edges switch on independently with probability p each step, so the aggregate is the backbone itself. As a real-world test we additionally use the empirical *SocioPatterns Hypertext 2009* face-to-face contact network [5] (113 attendees, 20 s resolution, ~ 59 h, binned into 5-min snapshots). Against each temporal run we compare two null models built from the same data: the *aggregated static* network (all edges present at every step) and a *time-shuffled* surrogate (contacts reassigned to random steps, destroying temporal order while preserving the aggregate and the total contact volume). A single seeded stochastic engine (Python + igraph) runs all four models on any of these sequences; results are averaged over 15–40 realisations.² Synthetic default: $N = 1500$, $T = 800$, $m = 1$, $\gamma = 2.1$, $\mu = 0.02$.

1.3 | Results

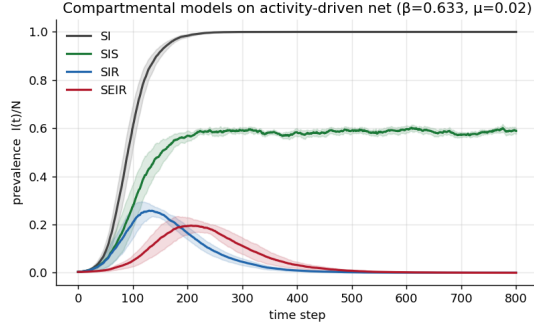
Figure 1.1(a) shows the four models on one AD network above threshold: SI saturates, SIS relaxes to an endemic plateau, and SIR/SEIR are single outbreaks, with SEIR delayed by the latent stage. Panel (b) locates the SIS threshold: the simulated prevalence departs from zero exactly at the activity-driven prediction β_c^{AD} of Eq. (1.1) (Table 1.1), far above the static-network value β_c^{HMF} . The time-shuffled curve coincides with the temporal one, confirming that in a memoryless activity-driven model the *ordering* of contacts is immaterial – the relevant effect is concurrency, not order. Panel (c) isolates that concurrency effect for SIR: collapsing the timeline into the aggregated static graph turns a slow, bounded outbreak (peak prevalence ≈ 0.18) into an explosive one (peak ≈ 0.98 , final size 1). Panel (d) exercises the models across backbone *structure*: with the same mean degree $\langle k \rangle = 4$, the scale-free BA backbone (hubs) yields a markedly larger outbreak than the homogeneous ER one (peak 0.12 vs. 0.06), while both remain far below their common aggregated-static limit. Treating a temporal network as static is therefore not a harmless approximation: it moves the epidemic threshold by nearly three orders of magnitude and grossly inflates the predicted outbreak.

Finally, Fig. 1.2 applies the same analysis to the empirical Hypertext 2009 contact network. The picture is qualitatively identical – aggregation inflates a bounded outbreak (final size 0.20) into a near-complete one (0.999) – but with one instructive difference from the synthetic case: *time-shuffling now lowers the outbreak* (final size $0.11 < 0.20$), because real contacts carry temporal correlations (bursts, group co-presence) that a memoryless activity-driven model lacks. Both the timing *and* the ordering of contacts matter in real data.

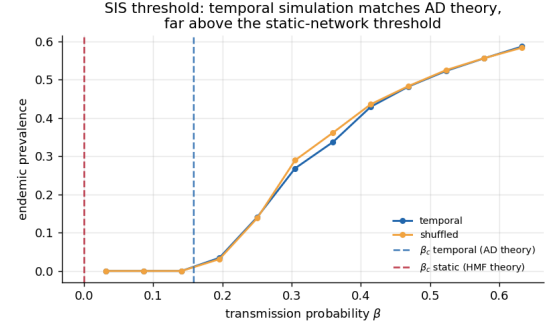
²Generators, engine, thresholds and output edge lists are in `code/task_22/` and `data/task_22/`; see Holme & Saramäki [4] for the temporal-network framework.

	β_c^{AD} (temporal)	β_c^{HMF} (static)	β_c^{QMF} (static)
$(\beta/\mu)_c$	7.91	9.5×10^{-3}	1.1×10^{-2}
β_c (at $\mu = 0.02$)	0.158	1.9×10^{-4}	2.2×10^{-4}

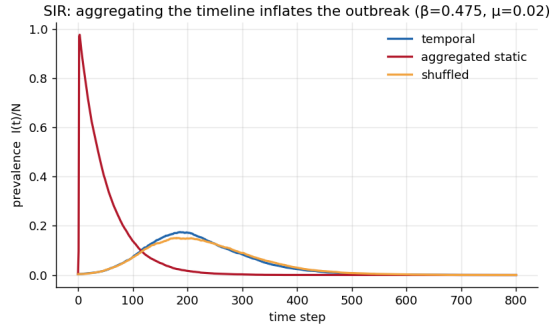
Table 1.1: Critical points for the default AD network ($\langle a \rangle = 0.041$, $\langle a^2 \rangle = 7.4 \times 10^{-3}$; aggregate $\langle k \rangle = 61$, $\langle k^2 \rangle = 6.4 \times 10^3$). The temporal threshold is $\sim 800\times$ the static one.



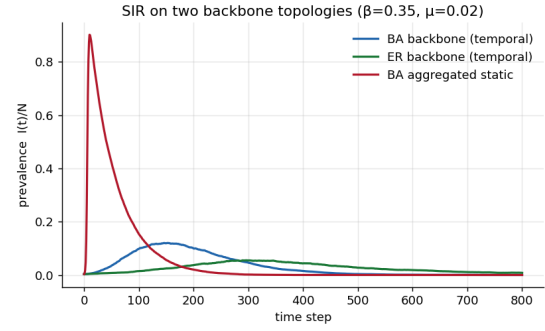
(a) Four models on the AD network



(b) SIS threshold vs. theory



(c) SIR: temporal vs. aggregated (AD)



(d) SIR: BA vs. ER backbones

Figure 1.1: Epidemics on temporal networks. (a) SI/SIS/SIR/SEIR prevalence on one activity-driven network ($\beta = 4\beta_c^{\text{AD}}$); bands are $\pm 1\sigma$ over realisations. (b) SIS endemic prevalence vs. β : the temporal simulation matches β_c^{AD} (dashed) and lies far above β_c^{HMF} ; time-shuffled $\approx$ temporal. (c) Aggregating the timeline (red) inflates the SIR outbreak relative to the temporal dynamics (blue). (d) Backbone structure at fixed $\langle k \rangle$: the scale-free BA network spreads more than the homogeneous ER one.

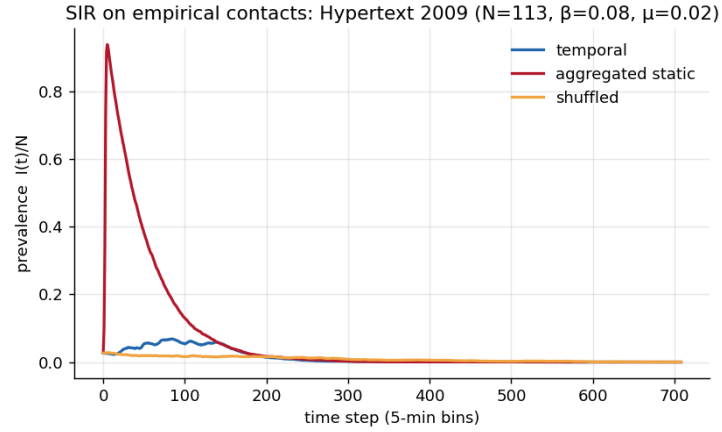


Figure 1.2: Application to empirical data: SIR on the SocioPatterns Hypertext 2009 face-to-face contact network. Aggregating the timeline (red) inflates the outbreak from a final size of 0.20 (temporal) to 0.999; unlike the memoryless synthetic models, time-shuffling (orange) *reduces* spreading (0.11), revealing temporal correlations in the real contacts.

2 | Monthly snapshots of the Internet AS-level network from BGP

Task leader(s): *Giancarlo Venturato*

2.1 | Data and network reconstruction

At the coarsest scale the Internet is a network of *Autonomous Systems* (ASes) that exchange reachability information through the Border Gateway Protocol (BGP). We reconstruct one AS-level snapshot per month from the routing-table dumps archived by the RIPE RIS collector `rrc00` [10], taking the first available `bview` table dump of each month (day 01 at 00:00 UTC when present; the exact file used is recorded in a manifest). The archive spans **November 1999 – April 2026**; 314 of the 318 calendar months yield a usable table (April–July 2000 are absent from the archive; in a few further months of 2000 and in June 2017 the first files are truncated, and the pipeline falls back to the largest remaining dump of the month). Three formats occur and are all handled: modern MRT `TABLE_DUMP/TABLE_DUMP_V2` records, parsed with `bgpdump`; the pre-2001 Zebra MRT dialect (many RIB entries packed into one record, 16-bit ASNs); and the gzipped GateD *text* tables of early 2000 — the latter two read by purpose-written parsers validated against neighbouring months.

Every `AS_PATH` is cleaned as follows: consecutive duplicate ASNs are collapsed (route prepending); malformed or empty paths are dropped; paths containing `AS_SETs` are dropped entirely, a documented simplification affecting a median 0.02% of entries per month; paths with fewer than two ASNs are dropped. Overall $\geq 96.4\%$ of RIB entries are retained in every month (median 99.9%; per-month statistics accompany the data). Each consecutive ASN pair of a retained path becomes an undirected edge in canonical (min, max) order, self-loops are discarded, and the integer edge weight counts the RIB entries in which the pair was observed that month; ignoring the weight column yields the unweighted simple graph used for all metrics below. ASNs are the integer AS numbers from the dumps, so node identity is consistent across snapshots.

An essential caveat: these graphs are *not* the physical Internet topology but the AS-level topology *as observed from a single collector*. An edge is visible only if some BGP peer of `rrc00` propagates a route through it, so peripheral (especially peer-to-peer) links are systematically under-sampled and all counts are lower bounds [6]. Moreover the collector’s own peer set grows in time (the monthly table has 4×10^5 RIB entries in 1999 and 5.5×10^7 in 2026), so part of any temporal trend reflects improving *visibility* rather than Internet structure; we point out below where this matters.

2.2 | Growth and heavy-tailed connectivity

Figure 2.1 summarises 26.5 years of evolution: the visible AS graph grows from 6 045 nodes and 10 634 edges (Nov. 1999) to 86 114 nodes and 239 705 edges (Apr. 2026) — factors of 14.2 and 22.5 respectively. Edges outpace nodes, so the network densifies, but only mildly: $\langle k \rangle$ rises from 3.5 to 5.6. The sharp $\langle k \rangle$ step around 2019–2021 coincides with a large expansion of the collector’s peer set and is largely a visibility effect, while the slow drift before it is consistent with genuine densification of the AS ecosystem. The graph is one connected component at all times ($\text{LCC} \geq 99.99\%$ of nodes in every month), as expected for a routing-derived view.

The degree distribution is heavy-tailed in every snapshot (Fig. 2.2a), the classical signature of the AS graph [3], with maximum degrees growing from $\sim 2.4 \times 10^3$ to $\sim 1.1 \times 10^4$. Discrete maximum-likelihood power-law fits [2, 1] give a strikingly stable exponent: $\hat{\alpha}$ stays within $[2.01, 2.21]$ for the whole series (typical $\hat{\sigma} \approx 0.03$), averaging 2.12 in the first five years and 2.10 in the last five (Fig. 2.2b) — the scale-free character of the AS topology is persistent while the network grows by more than an order of magnitude. Following [2] we also compared power-law and lognormal fits of each snapshot: the log-likelihood ratio mildly favours the lognormal in 282/314 months but is statistically inconclusive in 280 of them ($p > 0.1$), the well-known outcome that the AS degree tail cannot be distinguished from a lognormal on data of this size — “power law” is here a description of the regime $P(k) \sim k^{-2.1}$ over ~ 3 decades, not a model-selection claim.

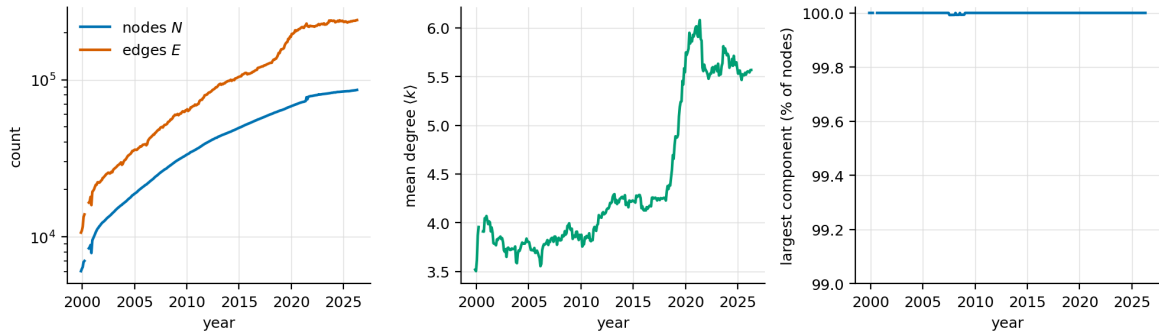


Figure 2.1: Growth of the monthly AS graph, Nov. 1999 – Apr. 2026. Node and edge counts (left, log scale), mean degree (centre; the 2019–2021 step is largely increased collector visibility), and largest-connected-component share (right). Line breaks mark the Apr.–Jul. 2000 archive gap.

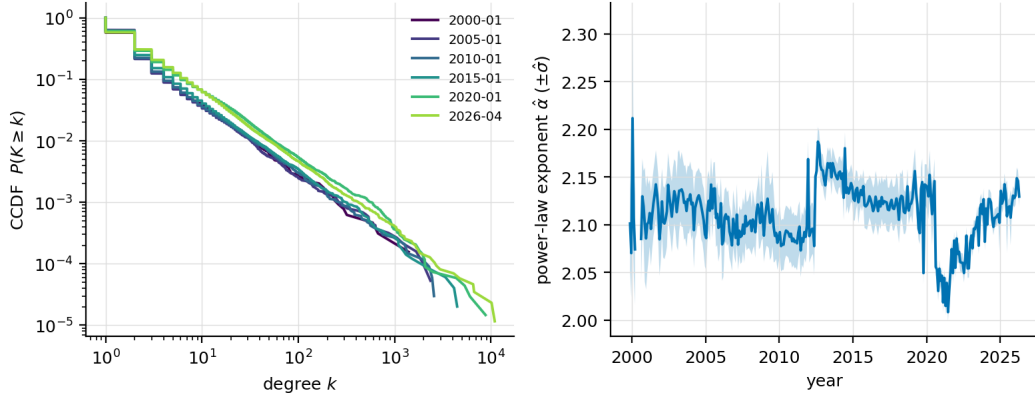


Figure 2.2: (a) Degree CCDF of six snapshots spanning the series: the tail $P(K \geq k) \sim k^{-1.1}$ ($\alpha \approx 2.1$) simply extends to larger k as the network grows. (b) Discrete MLE power-law exponent per month, with the fitted $\pm \hat{\sigma}$ band.

2.3 | Turnover, correlations and communities

Beneath the smooth aggregate growth the membership churns (Fig. 2.3): in a median month 1.3% of ASes are new and 0.7% of the previous month’s ASes disappear (node Jaccard overlap 0.98). Edges are far more volatile (Jaccard 0.90), partly through genuine rewiring of customer–provider relations and partly through fluctuating route visibility; both rates decay over the years as the network matures. The apex of the hierarchy, in contrast, is remarkably rigid: consecutive months share 95% of their top-20 highest-degree ASes on average, and hub identity changes only on multi-year timescales — the 1999 top of the list (AS701 UUNET, AS1239 Sprint, AS3561 Cable & Wireless, AS7018 AT&T) is gradually replaced by AS174 (Cogent), AS3356 (Level3/Lumen) and, at the top since the mid-2010s, AS6939 (Hurricane Electric). Hub *identity* is sticky even as hub degree grows by two orders of magnitude.

Degree–degree correlations are strongly disassortative throughout: $\langle k_{nn} \rangle(k) \sim k^\mu$ (Fig. 2.4a) with $\mu \approx -0.53$ around 2000, the value reported for the early Internet [7]. The exponent then drifts steadily to $\mu \approx -0.70$ by 2024–2026 (Fig. 2.4b): as the periphery grows, low-degree customers hang ever more exclusively off high-degree providers. Community structure found by Leiden modularity optimisation [11] (Fig. 2.4c,d) is significant at all times but evolves: Q rises to a plateau of ≈ 0.72 around 2010, then falls to ≈ 0.61 after 2019 while the number of communities triples (~ 30 to ~ 80). The post-2019 drop again partly tracks the visibility expansion — more visible interconnection (IXPs, content providers peering directly) genuinely flattens the hierarchy, and a better-instrumented collector sees more of it.

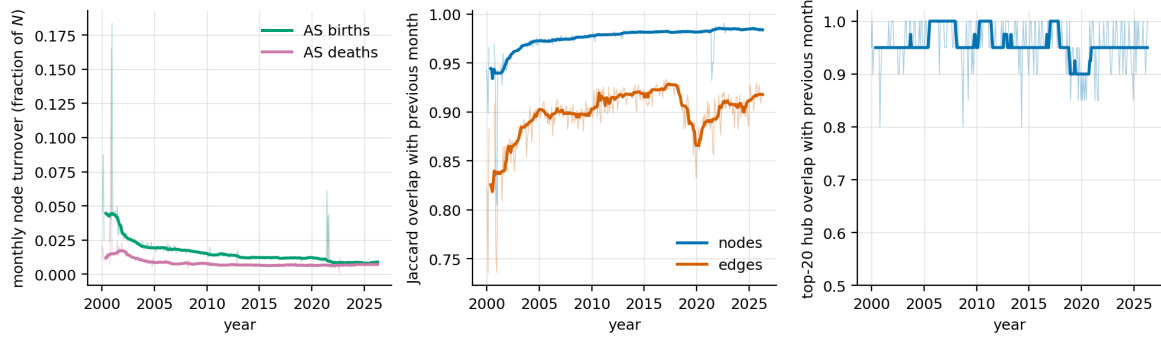


Figure 2.3: Monthly turnover (thin: raw; thick: 12-month rolling median). (a) AS birth/death rates. (b) Node and edge Jaccard overlap between consecutive months. (c) Top-20 hub set overlap between consecutive months.

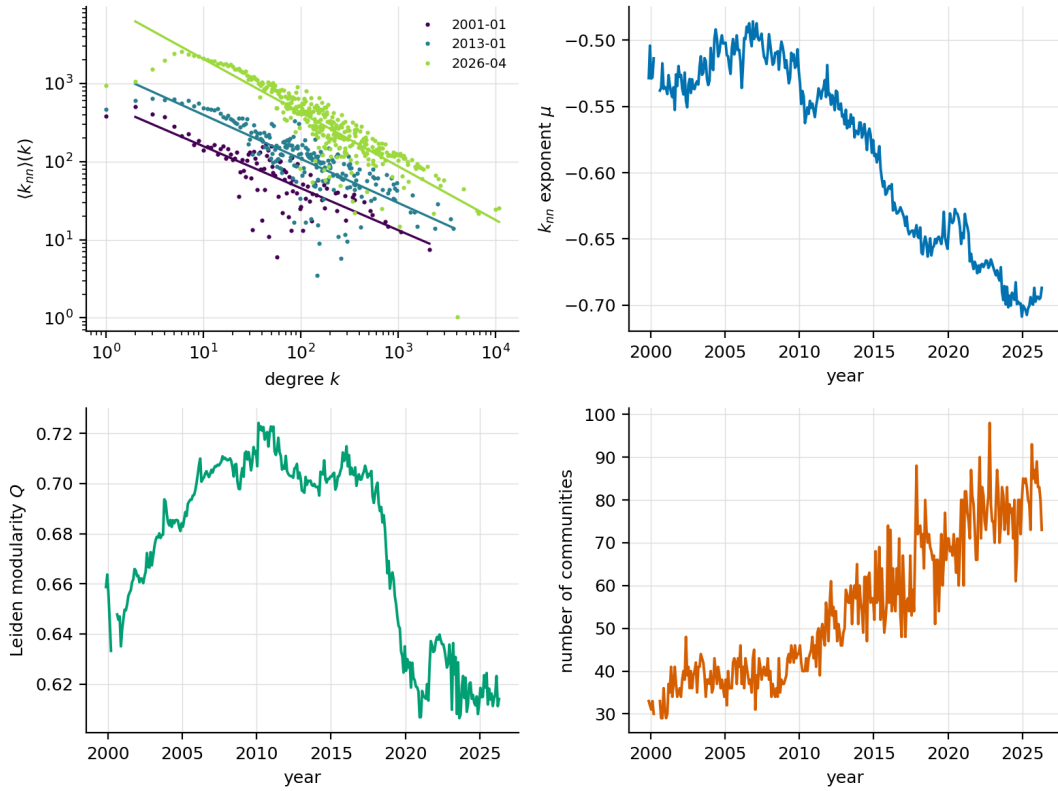


Figure 2.4: (a) $\langle k_{nn} \rangle(k)$ for three snapshots with power-law fits. (b) Disassortativity exponent μ per month. (c) Leiden modularity and (d) number of communities per month.

3 | Bibliography

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